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# Multiplicative information gating: decomposing news into novelty, materiality, and polarity for calibrated selective forecasting

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**Abstract:** A standard method for turning a news feed of events into a trading decision is to reduce each event to a single polarity and sum; this merges three price-impacting dimensions that work through different channels, viz. The sign of the news, its unexpectedness, and the amount to which it concerns value. We offer multiplicative information gating, which factors each event into polarity, novelty, and materiality, then multiplies them together; we set the resulting signal equal to near-zero whenever any factor is near-zero, so that that factor "vetoes" the signal. A LLM produces the three values in response to calibration-anchored prompts; we feed the signal to a calibrated probabilistic forecaster and then a conviction gate that may choose to sit out. We prove the product is the canonic combiner under per-axis multiplicative scaling, with veto and sign being the consistency constraints, and that the signal is bounded by its weakest factor. The method is market-agnostic and its performance environment ancillary; we instantiate it on exchange-traded equities with only measured numbers. With 621 resolved forecasts over a live append-only ledger, nominally 90% price confidence intervals achieved 70.7% actual coverage, showing that uncertainty as deployed in live-markets is not self-correcting; a three stage offline recalibration (which included fixing a stacking domain mismatch) reduced the expected calibration error from 0.35 to 0.05 without affecting the predictive power. There was essentially no signal in the next period direction of a single stock beyond drift (50.1% OOS; AUC 0.47), so ordering gave less value than selection. A conviction gate skipping 94% of cases pushed 30-day hit rates to 60.6% (95% CI 58.8 to 62.4) from 58.0% base and won four out of five test years; we provide an auditable, reusable template whose text stage and decision stage operate by a single principle of selectivity.

**Key words:** Calibration, selective prediction, natural language processing, large language models, financial forecasting, sentiment analysis

## 1. Introduction

A growing class of decision systems reads text and acts on it. News headlines, filings, social posts, support tickets, and clinical notes arrive as timestamped streams, and somewhere downstream a model turns them into a number that drives an action. The dominant recipe is disarmingly simple: score each item for sentiment polarity, average the scores, and feed the average to a predictor [1–3]. This works well enough in aggregate, and

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the financial literature has used it to show that media tone carries information about returns [1, 4, 5]. But the recipe rests on an assumption that breaks down on the events that matter most.

Sentiment polarity answers one question: is this good or bad? It is silent on two others that a careful reader weighs instinctively. Is the news actually new, or a restatement of something the market priced weeks ago? And does it bear on value, or is it cosmetic? A glowing but an unimportant stale press release can score around almost the same polarity as a regulatory filing that is quiet in tone recently, which might threaten the whole firm's existence, even though the impact of their price differs by huge orders of magnitude. When such items are summed and taken together, the loud and empty news are the ones that drown out the rare informative ones, because a common averaging out rewards volume over relevance in a system. The result is a final signal which looks reasonable in system backtests built on the same averaging method, and disappoints by performing poorly when a real position and value depends on it.

Now, we take the position that a textual event or maybe news, has at least three separable properties, wherein each property governs the price impact through a completely different channel. Let's define the three said properties. First is *Polarity*, which is the sign of the implied value change. Secondly, *Novelty*, which is how much genuinely new information any item carries, in the sense of surprise relative to what was already known by a system or model. Lastly, *Materiality*, which is how much sensitive fundamental value is to an event, independent of its sign value or freshness. A reader who collapses these three properties into one axis discards most of what makes the third item mentioned above worth acting on.

So, our proposal, called as multiplicative information gating, keeps the three axes of each property separately and combines them as a final product. So basically, the event signal is polarity times novelty times materiality. Because the factors are bounded and then at least two are non-negative, the product, in analogy, behaves like a soft logical AND, that is, a stale item (making its novelty near zero) or an immaterial one (whose materiality value is near zero) produces a signal near zero no matter how strong is its polarity value. A single weak factor invalidates the entire event. Unlike plain averaging, which has no such property, this is what lets a multiplicative signal keep a live position of an event from reacting to some noise that merely sounds important for the analysis.

But then, two practical questions might follow. First of all, who supplies the three scores? The hand-built dictionaries might capture polarity passably [3] but struggle with novelty and materiality values, which require real world knowledge and context. So, what we do is that we use a general-purpose instruction-tuned large language model, which is prompted with explicit numerical anchors so that its scores are reproducible and comparable across items [6, 7]. Secondly, once the gated signal enters a forecaster, how do we keep the system honest about what it does and does not know? So here the same gating principle reappears at the decision stage. A forecaster that is operating near the efficient-market limit [8, 9] is mostly wrong about direction; what it owes the user is not surely a confident guess on every name but is a calibrated probability and the discipline to abstain when conviction is low [10–12]. Calibration and selective abstention are, in this light, the decision-stage counterpart of the text-stage veto: each narrows what the system will commit to in return for being dependable on the rest.

We make the argument concrete on equities, where the data are public and the stakes are real, but the construction is market-agnostic and, we argue, domain-agnostic. The contributions are as follows.

1. We formalise multiplicative information gating: a decomposition of a textual event into polarity, novelty, and materiality, combined as a product, with a relevance-weighted aggregation to the entity level (Section 3). We prove a veto bound and show that, under five stated axioms whose substantive content is

per-axis multiplicative scaling, the product is the canonical combiner (Section 3.2).

2. We give a reproducible scoring protocol in which a large language model returns the three axes from calibration-anchored prompts, with a deterministic lexicon fallback and persistent memoisation so that a given item is scored once (Section 3.3).
3. We couple the gated text signal to a calibrated, selective forecaster, and we report measured results only: live-ledger interval coverage, a three-stage calibration that repairs a stacking domain mismatch, a verified accuracy ceiling, and a conviction gate evaluated walk-forward over an eight-year span, 2018 to 2026 (Sections 4).

Throughout, we are deliberate about what we can and cannot claim. The deployed ledger does not record per-event return labels, so we do not assert an isolated alpha for the text decomposition; its contribution is argued from its formal properties and its place in a pipeline whose end-to-end behaviour we do measure. We think this restraint is the right default for outward-facing systems, and Section 5 returns to it.

## 2. Related work

**Text and returns.** The idea that media content carries return-relevant information is well established. Tetlock [1] linked pessimistic media tone to downward pressure and subsequent reversal; Antweiler and Frank [4] studied message-board activity; Loughran and McDonald [3] showed that finance-specific dictionaries matter because general-purpose ones mislabel domain terms. Supervised text models tailored to returns improve on dictionaries [13], and pre-trained language models, including finance-tuned variants, now dominate sentiment extraction [14, 15]. Most of this work, however, treats the per-item output as a scalar polarity and aggregates by counting or averaging. The novelty and materiality of an item are typically handled, if at all, by ad hoc filters rather than as first-class, separately scored quantities. Boudoukh et al. [16] is a notable exception in spirit: identifying which news is relevant sharply changes the measured information content. We make relevance and surprise explicit, continuous, and multiplicative. Where Ke et al. [13] learn return-predictive word weights by supervising directly on realised returns, we deliberately operate in the complementary regime in which no per-item return label is available at scoring time: the three axes are judged zero-shot, and the veto property of Section 3.2 supplies a structural safety guarantee that a purely supervised weighting does not, namely that a stale or immaterial item cannot dominate however the labels happened to fall.

**Large language models as scorers.** Instruction-tuned language models can follow rubric-style prompts and emit structured numeric judgements [6, 17]. Recent work probes whether such models can read financial news and anticipate moves [7]. We do not ask the model to forecast; we ask it to *decompose*, which is a narrower and more checkable task, and we constrain it with numerical anchors so that two runs on the same item agree.

**Calibration.** Modern classifiers are often miscalibrated [18]. Post-hoc remedies include Platt scaling [19] and isotonic regression [20], with reliability assessed by expected calibration error and the Brier score [21–23]. Stacked generalisation [24] complicates this: a calibrator fitted on one stage’s scores and applied to another silently breaks, a failure we diagnose and repair in Section 3.4.

**Selective prediction.** Allowing a model to abstain is an old idea [10] with a modern theory [11, 12] and a distribution-free cousin in conformal prediction [25–27]. The financial-machine-learning literature, by contrast, is dominated by accuracy and area-under-the-curve reporting, which a long methodological line has shown to be fragile under backtest overfitting and multiple testing [28–30]. Deep and ensemble models for

markets are now standard [31–35], but few report whether their probabilities are trustworthy or whether they know when to stay out. Our decision stage is built around exactly those two questions.

### 3. Materials and methods

#### 3.1. The multiplicative information model

Let  $e$  denote a textual event (here, a news headline) about an entity (here, a listed equity). We assign  $e$  three scores:

$$s(e) \in [-1, 1] \text{ (polarity)}, \quad \nu(e) \in [0, 1] \text{ (novelty)}, \quad \mu(e) \in [0, 1] \text{ (materiality)}.$$

Polarity is the signed direction of the implied value change; novelty is the degree of genuinely new information, understood as surprise relative to the market’s prior; materiality is the sensitivity of fundamental value to the event, regardless of sign. The *event signal* is their product,

$$a(e) = s(e) \nu(e) \mu(e) \in [-1, 1]. \quad (1)$$

We define the *relevance* of an event as  $r(e) = \nu(e) \mu(e) \in [0, 1]$ , so that  $a(e) = s(e) r(e)$ : the event signal is its polarity scaled by how novel-and-material it is. To aggregate the events  $\mathcal{E}$  touching one entity into a single signal, we use a relevance-weighted mean restricted to events that clear a materiality floor  $\mu_0$ ,

$$A = \frac{\sum_{e \in \mathcal{E}: \mu(e) \geq \mu_0} r(e) a(e)}{\sum_{e \in \mathcal{E}: \mu(e) \geq \mu_0} r(e)} = \frac{\sum r(e)^2 s(e)}{\sum r(e)}. \quad (2)$$

Relevance is entered and used twice, one in the event signal and second in the aggregation weight, hence the power of 2, and so the cosmetic chatter is suppressed and an item that is both novel and material dominates quadratically. The floor  $\mu_0$  discards items below a minimum materiality before weighting; we set it as  $\mu_0 = 0.15$ .

The motivation is a first-order description of price response. If novelty proxies the magnitude of surprise relative to the market’s prior and materiality proxies the sensitivity of value to the underlying variable, then to first order the expected value move is the product of a sign, a sensitivity, and a surprise magnitude. Additive sentiment keeps only the sign and discards the coupling, which is why it over-reacts to loud, stale, or immaterial items. Equation (1) restores the coupling in its simplest form.

#### 3.2. Why the product, and what it guarantees

The multiplicative form is not arbitrary. It is the simplest combiner consistent with how a careful analyst treats the three axes.

**Proposition 1 (Veto bound)** *For every event  $e$ ,  $|a(e)| \leq \min\{|s(e)|, \nu(e), \mu(e)\}$ . In particular  $a(e) = 0$  whenever any single factor is zero, and the aggregate satisfies  $|A| \leq \max_e |s(e)|$ .*

**Proof** Each factor lies in  $[-1, 1]$  with  $\nu, \mu \geq 0$ , so  $|a(e)| = |s(e)| \nu(e) \mu(e)$  is a product of three numbers of modulus at most one; such a product cannot exceed the modulus of any one of them, which gives the per-event bound and the zero case. For the aggregate, (2) gives  $A = (\sum_e r(e)^2 s(e)) / (\sum_e r(e))$ , so

$$|A| \leq \frac{\sum_e r(e)^2 |s(e)|}{\sum_e r(e)} \leq \max_e |s(e)| \cdot \frac{\sum_e r(e)^2}{\sum_e r(e)} \leq \max_e |s(e)|,$$

where the last step uses  $\sum_e r(e)^2 \leq (\max_e r(e)) \sum_e r(e) \leq \sum_e r(e)$  because every  $r(e) \in [0, 1]$ . The weights  $r(e)^2 / \sum_j r(j)$  are non-negative but sub-stochastic, summing to  $\sum r^2 / \sum r \leq 1$ ; the aggregate is therefore a contraction toward zero of the polarities, never an amplification.  $\square$

Proposition 1 is the formal content of “a single weak factor vetoes the event.” A stale item ( $\nu \rightarrow 0$ ) or an immaterial one ( $\mu \rightarrow 0$ ) is capped near zero however extreme its polarity. No additive or convex combination of the three axes has this property, because a weighted sum  $\alpha s + \beta \nu + \gamma \mu$  is generally nonzero when one term vanishes. We state the design choice as a short characterisation.

**Proposition 2 (Characterisation)** *Consider combiners  $f(s, \nu, \mu)$  with  $s \in [-1, 1]$ ,  $\nu, \mu \in [0, 1]$  satisfying: (A0) normalisation,  $f(1, 1, 1) = 1$ ; (A1) sign fidelity,  $\text{sign } f = \text{sign } s$ ; (A2) veto,  $f = 0$  if any of  $|s|, \nu, \mu$  is 0; (A3) separable monotonicity,  $|f|$  is non-decreasing in each of  $\nu$  and  $\mu$  for fixed others, and  $f$  is non-decreasing in  $s$  for fixed  $\nu, \mu \geq 0$ ; and (A4) factor homogeneity, scaling any single argument by  $\lambda \in [0, 1]$  scales  $f$  by  $\lambda$ . Then  $f(s, \nu, \mu) = s \nu \mu$ .*

**Proof** [Proof sketch] By (A4) applied to each argument in turn,  $f(\lambda_1 s, \lambda_2 \nu, \lambda_3 \mu) = \lambda_1 \lambda_2 \lambda_3 f(s, \nu, \mu)$  for  $\lambda_i \in [0, 1]$ . Writing  $s = \text{sign}(s)|s|$  and taking  $\lambda_1 = |s|, \lambda_2 = \nu, \lambda_3 = \mu$  relative to the unit corner gives  $f(s, \nu, \mu) = |s| \nu \mu f(\text{sign}(s), 1, 1)$ . Sign fidelity (A1) forces  $f(\pm 1, 1, 1) = \pm c$  with  $c = f(1, 1, 1)$ , and the normalisation (A0) fixes  $c = 1$ . Hence  $f = s \nu \mu$ ; (A2) and (A3) are then automatically satisfied, so they are consistency requirements rather than the active constraints.  $\square$

We are explicit about where the force of the result lies. Homogeneity (A4) is the substantive axiom: it requires each axis to act on its own multiplicative scale, and once that and the normalisation (A0) are granted, the product follows in two lines. The veto and monotonicity conditions (A2, A3) do not by themselves single out the product—many functions satisfy them—so we do not claim the product is the *only* sensible combiner. We claim only that it is the canonical one once per-axis scaling is imposed, and that this scaling is exactly the property a position-taking system needs in order to treat freshness and relevance as preconditions for influence rather than as additive contributors.

### 3.3. Scoring the three axes with a language model

The decomposition is only as good as the three scores. We obtain them from a general-purpose instruction-tuned large language model (in our deployment, a compact production model, Claude Haiku 4.5), prompted as a quantitative analyst and constrained to return a JSON array of  $\{\nu, \mu, s\}$  triples, one per headline, in input order. Two design choices make the scores reproducible.

First, the prompt fixes *numerical anchors* for every axis, so the model maps qualitative judgements onto a shared scale rather than an idiosyncratic one. For novelty, 0.0 is an exact restatement of news already priced weeks ago, 0.2 a result that merely met consensus, 0.6 an unexpected but modest development, and 1.0 a sudden regulatory ban or fraud disclosure. For materiality, 0.0 is boilerplate, 0.4 a contract worth a few percent of revenue, 0.8 a transformative deal, and 1.0 an existential event. For polarity, the anchors run from  $-1.0$  (fraud allegation, pledge invocation) through 0.0 (a neutral reshuffle) to  $+1.0$  (a large earnings beat with raised guidance). Anchoring narrows the model’s effective output distribution and makes two runs on the same item agree closely, which matters because the product in (1) amplifies disagreement on any one axis.

Second, every scored item is memoised in a content-addressed store keyed by a hash of the entity and the normalised headline, so a given headline is scored once and reused thereafter; this makes a run deterministic

given its inputs and bounds cost. When the model is unavailable, a transparent lexicon fallback assigns polarity from curated bullish and bearish term lists and a conservative fixed novelty and materiality, flagged so that downstream consumers can discount it. The scorer is model-agnostic: any instruction-following model that respects the output contract can be substituted, and the anchors travel with the prompt rather than the weights.

### 3.4. From gated signal to calibrated probability

The aggregate  $A$  of (2) is one feature among several entering a hybrid forecaster that blends a gradient-boosted tree ensemble with a recurrent sequence model over price, momentum, trend, reversal, and volatility features to produce, for each name and horizon, a point forecast, a price interval, and a directional probability  $\hat{p}$ . We treat this forecaster as a black box; the object of study is not its architecture but its calibration and its willingness to abstain, which is where the two failures below live. A probability is only useful if it is calibrated: among names assigned  $\hat{p} \approx 0.7$ , about 70% should rise. We measure calibration by the expected calibration error over  $B$  equal-width bins,

$$\text{ECE} = \sum_{b=1}^B \frac{|G_b|}{N} |\bar{y}_b - \bar{p}_b|, \quad (3)$$

where  $\bar{p}_b$  and  $\bar{y}_b$  are the mean predicted probability and mean realised outcome in bin  $b$ , and by the Brier score  $\frac{1}{N} \sum_i (\hat{p}_i - y_i)^2$  [22].

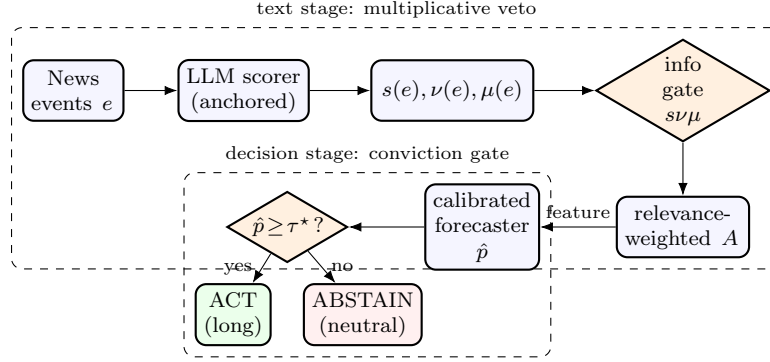
Our first deployed model reported a next-day up-probability near 0.80 while its realised accuracy sat near 0.51, an ECE of 0.345 and a Brier score worse than a constant 0.5 forecast. The cause was a stacking domain mismatch [24]: an isotonic calibrator had been fitted on the out-of-fold *average of the base learners* but then applied to the *stacked* prediction, a different distribution, and the final blend was never itself calibrated. The repair is a three-stage calibration. We (i) refit the isotonic map on the same base-average it was meant to correct, (ii) form the blend, and (iii) fit a final Platt scaler [19] on a held-out fold and apply it to both the test rows and the live row. Equation (3) is the diagnostic at each stage. The effect is reported in Section 4.2.

### 3.5. The conviction gate: abstention as decision-stage gating

Calibration makes  $\hat{p}$  honest; it does not make it skilful. If short-horizon single-name direction is near-random (Section 4.3), a calibrated probability will sit near the base rate almost everywhere, and acting on every name is pointless. The remedy mirrors the text-stage veto: act only when conviction clears a threshold, and abstain otherwise. For a horizon of  $H$  trading days we emit an UP call when the calibrated horizon probability satisfies  $\hat{p} \geq \tau^*$ , and NEUTRAL otherwise. The threshold is chosen on training folds as the smallest  $\tau$  whose fired bucket reaches a target precision (here 0.60) while still firing at least a minimum fraction of the time (here 5%):

$$\tau^* = \min \left\{ \tau : \text{prec}(\tau) \geq 0.60 \text{ and } \phi(\tau) \geq 0.05 \right\}, \quad (4)$$

where  $\text{prec}(\tau) = \Pr[y = 1 \mid \hat{p} \geq \tau]$  is the fired-bucket precision and  $\phi(\tau) = \Pr[\hat{p} \geq \tau]$  is the firing rate. (We reserve the word *coverage* for interval containment in Section 4.1; the firing rate  $\phi$  is its selective-prediction analogue.) The gate is intentionally one-sided: confident DOWN calls on single names are destroyed by the upward drift of equities (Section 4.4), so the system emits long-or-neutral signals only. As with the event veto,



**Figure 1.** Two gates, one principle. The text stage decomposes each event and vetoes stale or immaterial items through the product  $sv\mu$ ; the decision stage calibrates the forecaster’s probability and abstains unless conviction clears  $\tau^*$ . In both, breadth is given up for dependability. The schematic is conceptual.

the gate forgoes breadth for dependability, and the quantity worth reporting is therefore precision at a given firing rate rather than accuracy or area under the curve.

Figure 1 summarises the two gates in one pipeline.

### 3.6. Data, deployment, and evaluation protocol

The system is deployed on liquid National Stock Exchange of India (NSE) large-capitalisation equities. Two bodies of evidence are used, and we keep them strictly separate. The first is an *offline walk-forward* study on daily data from 2018 to 2026 for a 54-name universe, used to fit and evaluate the forecaster and the gate; all reported offline statistics are out-of-sample under an expanding window so that no future information enters a fitted fold. The second is a *live append-only ledger*: since April 2026 the deployed system has written every issued forecast (entity, timestamp, target date, point forecast, interval, nominal confidence, directional probability, horizon) to a database row that is later stamped with the realised price and outcome. The ledger is first-look only; a forecast is scored against the outcome recorded for its original target date and never retro-edited. This is the strongest evidence we have, because it is produced by the running system rather than reconstructed after the fact.

We report three kinds of measured result: interval coverage and calibration on the live ledger; the calibration repair and the accuracy ceiling from the offline study; and the selective gate’s precision at its firing rate, also offline and walk-forward. No quantity below is illustrative unless its caption says so.

## 4. Results

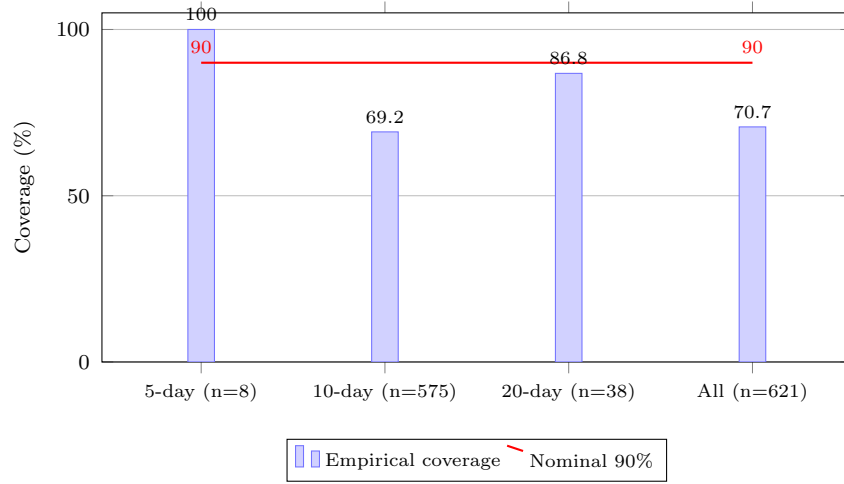
### 4.1. Deployed uncertainty does not self-correct

We begin with the live ledger because it sets the stakes. Between 19 April and 12 June 2026 the system issued and later resolved 621 price-interval forecasts on seven NSE large caps (ABB, HDFCBANK, ICICIBANK, INFY, RELIANCE, SBIN, TCS), each carrying a nominal 90% interval with a median half-width of 5.6% of price. If the intervals were well calibrated, about 90% should contain the realised price. They did not: empirical coverage was 70.7% overall, a 19.3 percentage-point shortfall. The bulk of the forecasts were at the ten-day horizon, where coverage was 69.2% over 575 resolved forecasts; the smaller twenty-day and five-day buckets covered 86.8% and 100% (Table 1, Figure 2). A deployed model’s stated confidence, in other words,

1 was systematically overstated, and nothing in the pipeline noticed on its own. This is the empirical reason  
 2 calibration and abstention cannot be afterthoughts.

**Table 1.** Live-ledger interval coverage by horizon (NSE large caps, 19 April–12 June 2026). All intervals carry a nominal 90% level.

| Horizon         | Resolved forecasts | Nominal coverage | Empirical coverage |
|-----------------|--------------------|------------------|--------------------|
| 5 trading days  | 8                  | 90.0%            | 100.0%             |
| 10 trading days | 575                | 90.0%            | 69.2%              |
| 20 trading days | 38                 | 90.0%            | 86.8%              |
| All             | 621                | 90.0%            | 70.7%              |



**Figure 2.** Empirical coverage of nominal 90% price intervals on the live ledger, by horizon. The deployed intervals are materially under-covered at the dominant ten-day horizon (575 of 621 forecasts).

#### 3 4.2. A stacking domain mismatch, and a three-stage repair

4 Offline overconfidence diagnoses are even more pointed: these two results, along with the one that preceded  
 5 them, are walk-forward analyses, not live-ledger numbers. The initial next-day directional head’s reported  
 6 probabilities lie concentrated near 0.80 while its actual accuracy falls around 0.51, resulting in  $ECE = 0.345$   
 7 and Brier Score  $\hat{\mu}$  0.366, worse than simply saying 0.5. Tracing the pipeline described in 3.4 indicated it as an  
 8 isotonic regression fitted on the base-learner averaged output was applied to the stacked model’s output, while  
 9 the final blend had no calibration. Its three-stage repair, fitting isotonic regression on the restricted domain  
 10 of its input, and finally a Platt scalar on a reserved fold, decreased ECE to 0.049 from 0.345, and brought  
 11 the reported up-probability close to a plausible 0.50 again. It’s critical that accuracy remains unchanged:  
 12 what changes is only what the model *claims to know about its confidence*. This episode stands out as a very  
 13 straightforward, widely transferable warning for any system utilizing stacking: any calibrator can only truly  
 14 learn a particular distribution and, stacking creates more than one.

### 4.3. Single-name direction is unpredictable beyond drift

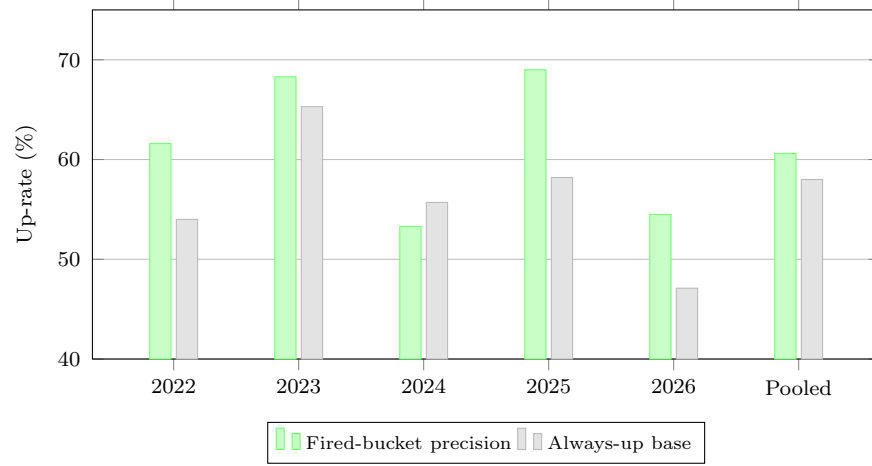
An honest calibration laid bare the truth that nice numbers had masked. We then pushed the probabilities down to base rate and asked more directly, does a robust single name directional edge remain in the short-term? The cross sectional model of five day direction on the entire universe generated 50.09% accuracy out of sample, with AUC of ranks 0.49, meaning the result was by chance. We validated the result on five data sets, using various learners and feature universes and conditioning on trend and/or regime and none could extract a consistent edge against the always-up base rate over a single day horizon. This is consistent with what efficient market theory predicts of liquid names [8, 9] and allows us to restate the issue as being selection not ranking: can we find those few events in which probability differs sufficiently from the base rate?

### 4.4. Selection recovers a small, stable edge

A longer horizon of 30 trading days (approximately 1.5 months) generates two small advantages: drift in equities (approximately 58% of the windows are up over a 30-day span) and some small idiosyncratic signal detected by a calibrated cross-sectional model on features on momentum, trend, reversal and volatility. On a walk-forward period of 2018-2026 on the 54-name universe (91,471 training rows and 50,272 out-of-sample rows), the confidence threshold (4) is determined to be  $\tau^* = 0.63$ . The fired UP bucket obtains 60.6% directional accuracy compared to an ever-up benchmark of 58.0%, an additional 2.6 percentage points; it fired on only 5.6% of the days, the model not participating on 94.4% of trading days. As the 60.6% precision arises from the pooled results of 2,840 calls, the 95% binomial interval is 58.8 to 62.4%, a one-tailed test versus the 58.0% benchmark yielded a  $z \approx 2.9$  (with  $p \approx 0.002$ ), so the pooled effect is statistically distinguishable from the benchmark. We have to resist overinterpreting this result: The year-to-year accuracy swings (from +10.8% in 2025 to -2.4% in 2024) are easily within the realm of sampling variation, since per year fired quantities were small (as illustrated in Table 2). This result can be best summarized as a small but robust tail-edge (not a powerful one), not an exceptional trading rule. The construction inherently targets precision on high-conviction tails and not ranking; as such, the area under the curve for this ranking rule is approximately 0.47, what the precision at the fired frequency is designed to capture and area-under-the-curve misses. For the 5 years in the test set, the gate produced accuracy above the benchmark on 4 out of the 5 years (Table 2 and Figure 3); the only negative year 2024 is displayed along the rest. By contrast, confident DOWN calls had out-of-sample accuracy below 10%, entirely decimated by drift, thus are never fired by the model.

**Table 2.** Walk-forward selective signal, by test year. Precision is the realised up-rate of the fired high-conviction bucket; base is the always-up rate that year; fires is the number of high-conviction calls that year.

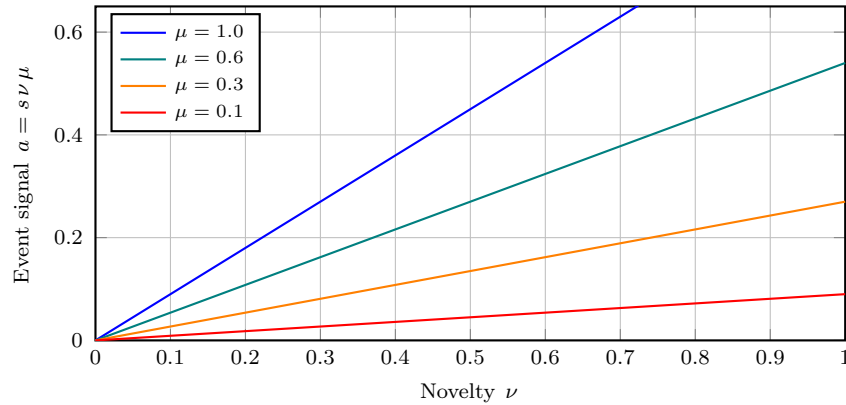
| Test year | Fires | Fired-bucket precision | Always-up base | Edge (pp) |
|-----------|-------|------------------------|----------------|-----------|
| 2022      | 365   | 61.6%                  | 54.0%          | +7.6      |
| 2023      | 290   | 68.3%                  | 65.3%          | +3.0      |
| 2024      | 932   | 53.3%                  | 55.7%          | -2.4      |
| 2025      | 822   | 69.0%                  | 58.2%          | +10.8     |
| 2026      | 431   | 54.5%                  | 47.1%          | +7.4      |
| Pooled    | 2,840 | 60.6%                  | 58.0%          | +2.6      |



**Figure 3.** Selective signal versus its base rate by test year. The conviction gate beats the always-up base in four of five years and pooled, at about a 5.6% firing rate.

#### 4.5. The veto in action

Since the per-event returns are not stored in the deployed ledger, we can not evaluate the text stage as an isolated alpha; we are very explicit about this limitation in Section 5. But what we can surely show is that the decomposition behaves as designed and intended. Figure 4 plots the event signal  $a = s\nu\mu$  as a relative function of the novelty value for several materiality levels at a fixed positive polarity. As seen, the product collapses toward zero as either the novelty or materiality falls, so a strongly worded event but a stale one or an immaterial headline cannot move the aggregate function value that the protection Proposition 1 guarantees, and which the protection additive sentiment cannot offer. Table 3 makes the same point on a small illustrative set of headlines. Both a shiny but routine in-line result and an emphatic but cosmetic announcement are gated down, while a quieter disclosure which is, in reality, genuinely new and so it's material dominates despite its more moderate polarity. The example is illustrative of the scoring behaviour and carries no performance claim as such.



**Figure 4.** Response of the event signal to novelty at fixed polarity  $s = 0.9$ , for four materiality levels. Low novelty or low materiality vetoes the signal regardless of polarity.

**Table 3.** Illustrative decomposition of four headlines for one entity. Scores follow the anchors of Section 3.3; the example shows the gating behaviour and is not a performance measurement.

| Headline (paraphrased)                                | Polarity $s$ | Novelty $\nu$ | Material. $\mu$ | Signal $a$ |
|-------------------------------------------------------|--------------|---------------|-----------------|------------|
| Quarterly profit in line with consensus, up-beat tone | +0.30        | 0.20          | 0.30            | +0.018     |
| Company reiterates previously announced buyback       | +0.55        | 0.10          | 0.40            | +0.022     |
| Unexpected regulator notice on a core product line    | -0.55        | 0.75          | 0.80            | -0.330     |
| Routine management reshuffle, no strategic change     | +0.05        | 0.40          | 0.15            | +0.003     |

## 5. Discussion

Every result here turns on the same trade-off. At the text stage, the multiplicative veto refuses to let stale or immaterial items contribute, and many headlines are deliberately near-zeroed. At the decision stage, the conviction gate stays out on the great majority of names. The system thus surrenders any pretence of commenting on everything, and earns in exchange the right to be believed on the little it does flag. We think this is the correct posture for any model whose outputs are read literally by people with money or safety at stake.

**Why multiplicative rather than additive.** The veto bound of Proposition 1 is not a technicality; it is the whole point. Additive sentiment treats a loud restatement and a quiet revelation as comparable because it scores only direction and magnitude of tone. The product makes relevance a precondition for influence. We cannot attribute a standalone return to this choice on the present evidence, but the design consequence is clear and testable wherever per-event labels exist: under the product, the events that move the aggregate are those an analyst would flag as new and consequential, and no others.

**Generality.** Nothing in Section 3 is specific to any of the equities. Any stream of timestamped textual events or maybe news that drives a decision, all have the same three axes, that is, one, a direction of implied effect, second, a degree of surprise, and third, a degree of consequence. All of the following, sorting of clinical alerts, prioritisation of security or operation tickets, and moderation queues, when collapsed to a single urgency or sentiment score, suffer the same failure, and all could use a veto that suppresses what is in reality, loud yet stale or even forceful yet immaterial. Then, the decision-stage discipline is equally portable, calibrate the probability, then at a chosen fire rate, refrain below a conviction threshold tuned for precision. We chose finance as a domain, because its outcomes are public and unforgiving in true nature, and we framed the study as a market-agnostic case so as not to lean on results peculiar to a single venue as such.

**Threats to validity and honest limits.** Four limits deserve emphasis. First, the language-model scores are judgements, not measurements; anchoring tightens them but does not make novelty and materiality objective, and a different model or a drifting prompt could shift the scale. The deterministic fallback and the memoisation bound this variance without eliminating it. Second, the live ledger is young, two months and seven names, so its coverage shortfall, though large and consistent at the dominant horizon, is a snapshot, and we report it as a current state rather than a stationary law. Third, and the limitation we weight most heavily, we do not have per-event return labels in deployment, so the marginal value of the decomposition over plain polarity is argued from its formal properties and its place in the measured pipeline rather than from an isolated backtest. We would rather say this plainly than manufacture a clean number from a reconstruction,

which is the sort of exercise that invites the backtest-overfitting critique [28, 29] we take seriously. Fourth, the directional-probability sample on the live ledger is still small, only 28 resolved next-period calls, which is why our live claim rests on the 621-forecast interval-coverage evidence and not on a live reliability diagram.

**Relation to selective and conformal prediction.** The decision gate is an instance of selective classification [10, 12]; what is new here is pairing it with a calibrated probability whose calibration was itself repaired, and reading both as the same gating principle that governs the text stage. Distribution-free interval methods [26, 27] are a natural next step for the price intervals whose under-coverage we documented in Section 4.1; adaptive conformal inference in particular is built for the kind of non-stationary coverage failure we observe, and integrating it is left for future work.

## 6. Conclusion

Turning text into decisions should not begin by collapsing every event to a single sentiment number. A textual event carries direction, surprise, and consequence, and these need not move together. Taking their product instead of their sum builds a signal in which the weakest axis caps the rest, the safeguard that keeps a live decision from chasing loud noise. We proved that the product is the canonical combiner once per-axis multiplicative scaling is imposed, and bounded the aggregate by its strongest polarity. The same gating logic belongs at the decision stage. On a live deployment ledger, nominal 90% intervals covered only 70.7% of outcomes, so a forecaster has to be calibrated and willing to abstain. An offline three-stage recalibration that first repaired a stacking domain mismatch cut expected calibration error from 0.35 to 0.05 without touching accuracy; the corrected probabilities then exposed that short-horizon single-name direction is unpredictable beyond drift; and a conviction gate that stays out on about 94% of names raised the thirty-day hit rate to 60.6% against a 58.0% base, in four of five test years. What ties the two stages together is one rule: give up breadth where the signal cannot be trusted, and commit only where it can. Per-event labelled evaluation of the decomposition, and adaptive conformal intervals for the price forecasts, are the clear next steps.

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## Declaration of generative AI and AI-assisted technologies

The methodology of this study uses a large language model as a measurement instrument: an instruction-tuned model (Claude Haiku 4.5, Anthropic) scores news headlines for novelty, materiality, and polarity from the anchored prompt described in Section 3.3; this use is part of the system under study and all such outputs were validated against the deterministic fallback and persisted for audit. Separately, during the preparation of this manuscript the authors used a large language model solely to improve language and readability of author-written drafts. No generative tool was used to produce data, results, figures, or interpretations; all quantitative findings derive from the authors' system and its measured ledger. After using these tools the authors reviewed and edited the content and take full responsibility for the publication.

## Data availability statement

The live forecast ledger and the offline walk-forward statistics underlying Tables 1–2 and Figures 2–3 can be made available by the corresponding author on reasonable request, subject to the platform’s terms. Source price data are from public NSE end-of-day records.

## Conflict of interest

The authors declare that they have no conflict of interest.

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